

Finite One-Point Rigidification for Lipschitz-Free Spaces

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Source And Open Problem

The source paper is Cuth–Doucha–Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266 [1]. In Section 8.2 it asks:

EXAMPLE 8.1.0 ([1]). Therefore we get that $\dim(\mathfrak{q}) = 1 + 21 + 2$ and we are done.

8.2. **Problems.** We find the following question the most pressing.

Question 1. Does there exist a Lipschitz-free rigid metric space $\mathcal{M}$ such that $\bigcup E_{ext}(\mathcal{M})$ is not dense in $\mathcal{M}$?

A very interesting answer to the previous question would be answering positively the next question. However, even a negative answer to the next question would be interesting.

Question 2. Is $\mathbb{R}^d$, for $d \geq 2$, Lipschitz-free rigid? Either if it is equipped with the euclidean or with the Manhattan distance.

Question 3. Does every metric space isometrically embed into a Lipschitz-free rigid space that contains only one additional point?

REFERENCES

[1] M. CUTH, M. DOUCHA, I. G. GARCÍA-LÓPEZ, AND A. ZIMMER, *Geometry and isometry*, arXiv:2402.08266.

We prove a finite version of Question 3: every finite metric space admits a Lipschitz-free rigid one-point extension.

Proof Intuition

The source paper shows that surjective linear isometries of weak Prague Lipschitz-free spaces are encoded by bijections of preserved-extreme molecules satisfying three combinatorial and metric axioms. We exploit this encoding directly.

Given a finite metric space M , add a point p whose distances $r_x = d(p, x)$ are all very close to a large constant, so the old metric is preserved and every radial pair $\{p, x\}$ is preserved extreme. Choose the finitely many numbers r_x algebraically independent over the field generated by the old distances. Then each old preserved edge $\{x, y\}$ forms a triangle

$$\{p, x\}, \quad \{x, y\}, \quad \{y, p\}.$$

On this triangle, the source ratio axiom cannot send a radial edge to an old edge, or to a different radial edge, without creating an algebraic relation among the r_x 's. Therefore every preserved-extreme edge is fixed as an undirected edge. The directed-cycle axiom then forces all edge orientations to be either preserved globally or reversed globally. Thus the only linear isometries of the free space are I and $-I$, which are rigid in the required sense.

Definitions Used

For a finite metric space X , let $G_{\text{ext}}(X)$ be the undirected graph on X whose edges are the unordered pairs $\{a, b\}$ such that the molecule $m_{a,b}$ is a preserved extreme point of the unit ball of $\mathcal{F}(X)$. By the source paper’s preserved-extreme criterion, in a finite metric space this is equivalent to

$$[a, b] = \{a, b\},$$

that is, no third point lies between a and b .

Lemma 1 (Finite preserved-extreme graph is connected). *If M is a finite metric space with at least two points, then $G_{\text{ext}}(M)$ is connected and has vertex set M .*

Proof. Fix distinct $x, y \in M$. We prove by induction over the finite set of distances in M that x and y are connected by a path in $G_{\text{ext}}(M)$. If $[x, y] = \{x, y\}$, then $\{x, y\}$ is an edge. Otherwise there is $z \in M \setminus \{x, y\}$ with

$$d(x, y) = d(x, z) + d(z, y).$$

Both $d(x, z)$ and $d(z, y)$ are strictly smaller than $d(x, y)$, so the induction hypothesis gives preserved-extreme paths from x to z and from z to y . Concatenating them gives a path from x to y . Since x, y were arbitrary, the graph is connected and every point is incident to an edge. $\square$

Result

Theorem 1 (Finite one-point rigidification). *Let M be a finite metric space. There is a metric space $M^+ = M \cup \{p\}$, obtained by adding one point and leaving all distances inside M unchanged, such that M^+ is Lipschitz-free rigid.*

Proof. If $|M| \leq 1$, add one point at distance 1. The resulting free space is one-dimensional, so its only linear isometries are I and $-I$, and it is Lipschitz-free rigid. Hence assume $|M| \geq 2$.

Let

$$D = \text{diam}(M), \quad \Delta = \min\{d(x, y) : x \neq y \in M\} > 0.$$

Let K be the subfield of $\mathbb{R}$ generated over $\mathbb{Q}$ by the finitely many old distances $d(x, y)$, $x, y \in M$. Choose $R > D$ and choose numbers

$$r_x \in (R, R + \Delta/4) \quad (x \in M)$$

which are algebraically independent over K . This is possible because K is countable and the algebraically dependent tuples form a countable union of proper algebraic zero sets in the open box $(R, R + \Delta/4)^M$.

Define $M^+ = M \cup \{p\}$ by keeping the original metric on M and setting

$$d(p, x) = r_x, \quad x \in M.$$

This is a metric. Indeed, for $x \neq y \in M$,

$$|r_x - r_y| < \Delta/4 < d(x, y)$$

and

$$d(x, y) \leq D < 2R < r_x + r_y.$$

We next compute the preserved-extreme graph of M^+ . Since M^+ is finite, it suffices to check metric betweenness. If $\{x, y\} \subset M$, then the new point p is not between x and y , because $r_x + r_y > d(x, y)$. Thus an old pair is preserved extreme in M^+ exactly when it was preserved extreme in M . For a radial pair $\{p, x\}$, no old point $y \neq x$ lies between p and x , because

$$r_x < r_y + d(x, y) \quad \text{and} \quad r_y < r_x + d(x, y)$$

follow from $|r_x - r_y| < d(x, y)$. Hence every radial pair $\{p, x\}$ is preserved extreme. Therefore

$$G_{\text{ext}}(M^+) = G_{\text{ext}}(M) \cup \{\{p, x\} : x \in M\}.$$

In particular M^+ is a finite weak Prague space, because the radial star makes the preserved-extreme graph connected and all vertices are present.

Let E denote the directed preserved-extreme edge set of M^+ . By the source paper's correspondence theorem for weak Prague spaces [1, Theorem 3.4], every surjective linear isometry of $\mathcal{F}(M^+)$ induces a symmetric bijection $\sigma : E \rightarrow E$ satisfying the source conditions (SA), (SB), and (SC). It is enough to prove that every such σ is either the identity on directed edges or the global reversal $e \mapsto -e$.

Write $s_x = \{p, x\}$ for the undirected radial edge at x . Let $e = \{x, y\}$ be an old preserved-extreme edge. The three undirected edges

$$C_e = \{s_x, s_y, e\}$$

form a simple cycle in $G_{\text{ext}}(M^+)$. The cycle condition sends C_e to a three-edge simple cycle. We claim that the induced undirected edge map fixes C_e edge by edge.

The old edge lengths all belong to K , while the radial lengths are the algebraically independent numbers r_z . On the directed version of C_e , the source ratio axiom says that the three quotients

$$\frac{\text{source edge length}}{\text{target edge length}}$$

are equal. The target three-cycle cannot be an entirely old triangle, since then this common quotient would be in K when computed from the old edge, but would also equal r_x/a for some $a \in K$, an algebraic relation over K .

Thus the target is another cone triangle with two radial lengths and one old length. The old edge e must map to the old edge of that target triangle: if an old length $a \in K$ mapped to a radial length r_u while a radial length r_x mapped to an old length $b \in K$, equality of the quotients would give $ab = r_u r_x$, again a nonzero algebraic relation over K . Hence the old edge maps to an old edge, so the common quotient lies in K . If r_x mapped to r_u , equality of quotients gives $r_x = \lambda r_u$ with $\lambda \in K$. Algebraic independence forces $u = x$ and $\lambda = 1$. Similarly r_y maps to r_y , and then the old target edge is again $\{x, y\}$. This proves that s_x , s_y , and e are fixed as undirected edges.

Since every old preserved-extreme edge e was arbitrary, all old undirected edges are fixed. By Lemma 1, every $x \in M$ is incident to an old preserved-extreme edge, so every radial edge s_x is fixed as well.

It remains only to control orientations. For each directed edge $a \in E$, the undirected edge underlying a is fixed, so $\sigma(a)$ is either a or $-a$. Orient an old edge $e = \{x, y\}$ as the directed triangle

$$(p, x), \quad (x, y), \quad (y, p).$$

The image of this directed three-cycle is again a directed three-cycle. A direct check on the three fixed undirected sides shows that this is possible only when the orientation signs on (p, x) , (x, y) ,

and (p, y) are all the same. Since $G_{\text{ext}}(M)$ is connected by Lemma 1, these equalities propagate through the old graph, giving a single sign $\varepsilon \in \{\pm 1\}$ such that

$$\sigma(a) = \varepsilon a, \quad a \in E.$$

Consequently the associated linear isometry T satisfies $T(m_a) = m_a$ for every $a \in E$, or $T(m_a) = -m_a$ for every $a \in E$. The radial molecules $m_{p,x}$, $x \in M$, span $\mathcal{F}(M^+)$, so $T = I$ or $T = -I$. These are induced by the identity metric dilation of M^+ , with sign $+1$ or -1 . Hence M^+ is Lipschitz-free rigid. $\square$

Verification And Limitations

This proves Question 3 for all finite metric spaces. It strictly strengthens the earlier finite 2-connected preserved-extreme graph packet from this run, because no connectivity hypothesis on $G_{\text{ext}}(M)$ remains.

The theorem does not settle the full question for arbitrary infinite metric spaces. The proof uses the finite distance field and a finite algebraic-independence choice. Extending this to infinite spaces would require a different coding or compactness argument, especially when the set of old distances is already too large for real algebraic independence over its generated field to be useful.

Novelty check: local run indexes had no answer to arXiv:2402.08266 before this work, apart from the earlier partial packet produced in this run. Web searches on 2026-07-03 for exact and close phrases such as “Lipschitz-free rigid one additional point”, “finite metric space Lipschitz-free rigid one”, and “one-point Lipschitz-free rigid finite metric” found the source paper but no later finite one-point solution.

The scratch script `code/cone_admissible_search.py` enumerates small graph-metric cone examples and found no non-spatial admissible symmetry satisfying the source norm identities. It is included only as sanity evidence; the proof above is not computational.

Human Review Recommendation

Reviewers should check two points carefully:

1. the algebraic-independence argument showing that every cone triangle is fixed edge by edge under the source ratio axiom;
2. the directed-cycle orientation argument forcing one global sign.

References

- [1] M. Cuth, M. Doucha, and T. Titkos, *Isometries of Lipschitz-free Banach spaces*, arXiv:2402.08266, 2024. <https://arxiv.org/abs/2402.08266>